SECTIONS OF AN ARBITRARY DIRECT SUM OVER A QUASI-COMPACT OPEN ARE THE DIRECT SUM OF THE SECTIONS

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The direct sum of a family of sheaves of modules and the description of its sections by the sheafification of the direct sum presheaf. The author's section on the abelian category of sheaves of modules, in which the counted statement occurs. Supplies the map whose bijectivity is asserted.
- The equality between the sections of a direct sum sheaf and the direct sum of the sections for a finite index set. Invoked at the end of the counted proof (source lines 532-534) as a pointer to an earlier lemma of this chapter. Reduces the statement to the observation that only finitely many summands can occur.

Source reference. Original lemma lemma-limits-colimits: sources/stacks-modules/pinned-original-source.tex, source lines 534–534. Navigation pointer only: the counted text refers to the Stacks item lemma-limits-colimits, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Lemma 0.1. *Let $(X, \mathcal{O}_X)$ be a ringed space. Let I be a set. For $i \in I$, let $\mathcal{F}_i$ be a sheaf of $\mathcal{O}_X$-modules. For $U \subset X$ quasi-compact open the map*

$$\bigoplus\nolimits_{i \in I} \mathcal{F}_i(U) \longrightarrow \left(\bigoplus\nolimits_{i \in I} \mathcal{F}_i\right)(U)$$

is bijective.

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Proof. If s is an element of the right hand side, then there exists an open covering $U = \bigcup_{j \in J} U_j$ such that $s|_{U_j}$ is a finite sum $\sum_{i \in I_j} s_{ji}$ with $s_{ji} \in \mathcal{F}_i(U_j)$. Because U is quasi-compact we may assume that the covering is finite, i.e., that J is finite. Then $I' = \bigcup_{j \in J} I_j$ is a finite subset of I. Clearly, s is a section of the subsheaf $\bigoplus_{i \in I'} \mathcal{F}_i$. The result follows from the fact that for a finite direct sum sheafification is not needed, see Lemma lemma-limits-colimits above. $\square$